



MAKERERE UNIVERSITY BIOMETRIC ATTENDANCE MANAGEMENT SYSTEM

(Mak-BAMS)

GUIDELINES AND PROCEDURES

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1. THE PHYSICAL IDENTITY AND ACCESS MANAGEMENT FRAMEWORK OF MAKERERE UNIVERSITY (Mak-PIAM).

The Makerere Biometric Attendance Management System (Mak-BAMS) has been implemented as a modular component in the Makerere University Physical Identity and Access Management Framework (Mak-PIAM) which aims to address;

1.1. Security for University Staff, Students, Visitors and properties

Physical Access control Systems (PACS) have been implemented at university gates and premises/buildings. These comprise of;

- i. A Traffic Management & Automated Access Barrier System for monitoring motor-vehicles – at established entry and exit gates.
- ii. Automated Pedestrian Gates/Access Barrier System for Pedestrians (Staff, Students and Visitors) at established entry and exit gates.
- iii. Electronic Door Systems for Staff, Students and Visitors entering and exiting university premises/buildings.
- iv. An Enterprise Video Surveillance System/ CCTV network for real time digital surveillance addressing all security concerns in the University.

1.2. Attendance Management & Payroll Verification for staff

A Biometric Attendance Management System (Mak-BAMS) has been implemented under the Mak-eHRMS and staff are able to view their daily hours worked. The Directorate of human Resources is also able to monitor staff attendance in real time.

1.3. Lecture Attendance Management for students

A Mobile-Based Attendance Management System (Mak-SAMS) has been implemented and integrated with ACMIS and Mak-eHRMS to provide real time student lecture Attendance Tracking.

1.4. Performance Management for staff

An appraisal tool has been implemented under the Makerere University electronic Human Resource Management System (Mak-eHRMS) enabling staff to log tasks assigned to them, appraise staff performance against agreed deliverables for a given period.

1.5. Secure access to all university electronic resources and Information Systems

Logical Access Control Systems (LACS) such as Single Sign-On (SSO) have been implemented to ensure unique a lifeline tracking of both staff and students via the email-based authentication.

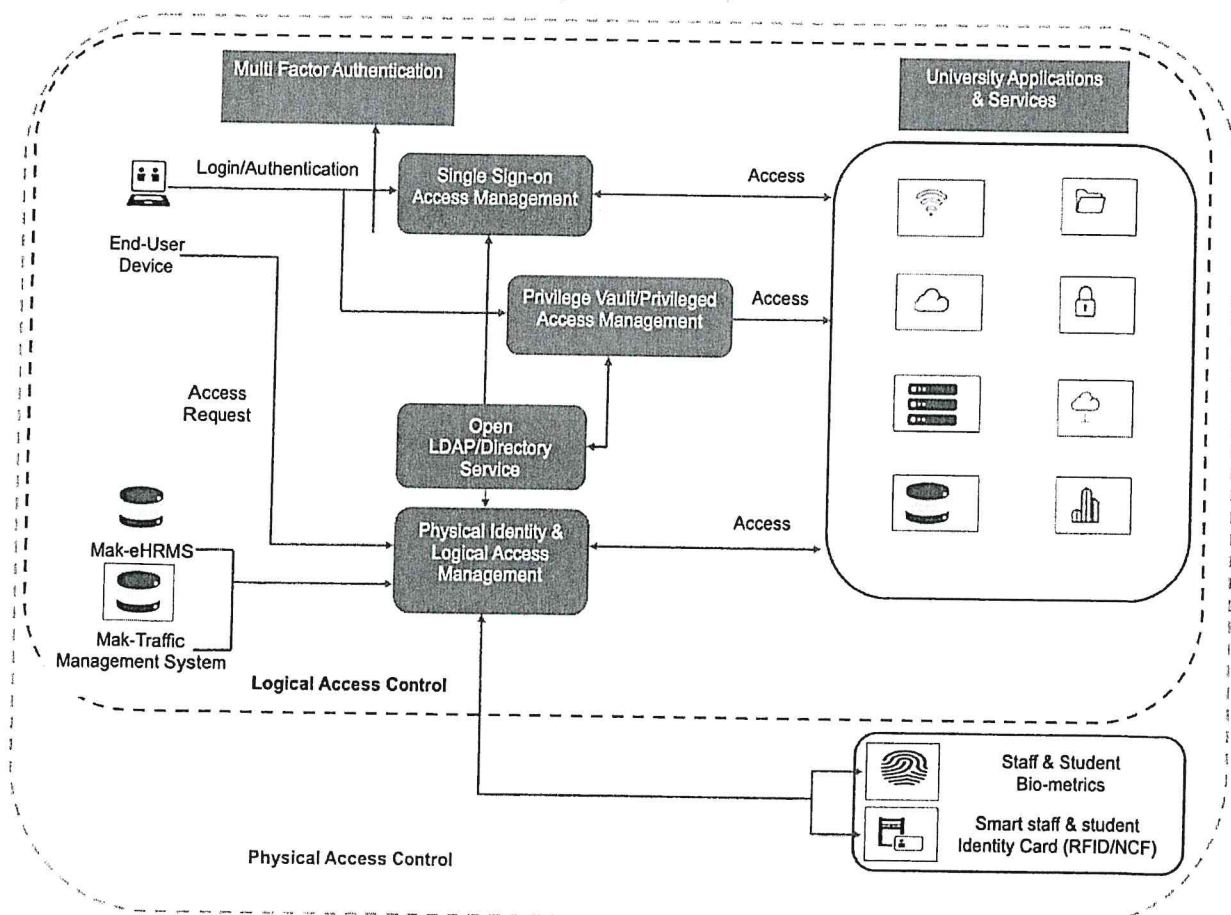
1.6. Electronic Quality Assurance (eQA) of both physical and virtual Teaching and Learning

The Student Evaluation of Course and Teaching system (SECAT) has been integrated with the Academic Management Information System (ACMIS), the ACMIS Time-Table Module, the Student Attendance Management System, the electronic Human Resource Management System (eHRMS) and the Makerere University Electronic Learning Environment (MUELE) system providing real time tracking of the staff teaching loads.

1.7. Graduate Research Supervision

The electronic graduate supervision system under the Research Information Management System (RIMS) has been implemented to provide a real time tracking of the graduate students' research processes.

2. THE ARCHITECTURE OF THE PHYSICAL IDENTITY AND ACCESS MANAGEMENT FRAMEWORK (Mak-PIAM).



3. THE ARCHITECTURE OF THE BIOMETRIC CLOCK-IN SYSTEM

3.1. Introduction

This section explains the design of the Mak-BAMS clock-in and out system, and highlights the proposed implementation modalities (see Appendix B for detailed technical description).

3.2. Mak-BAMS Architecture

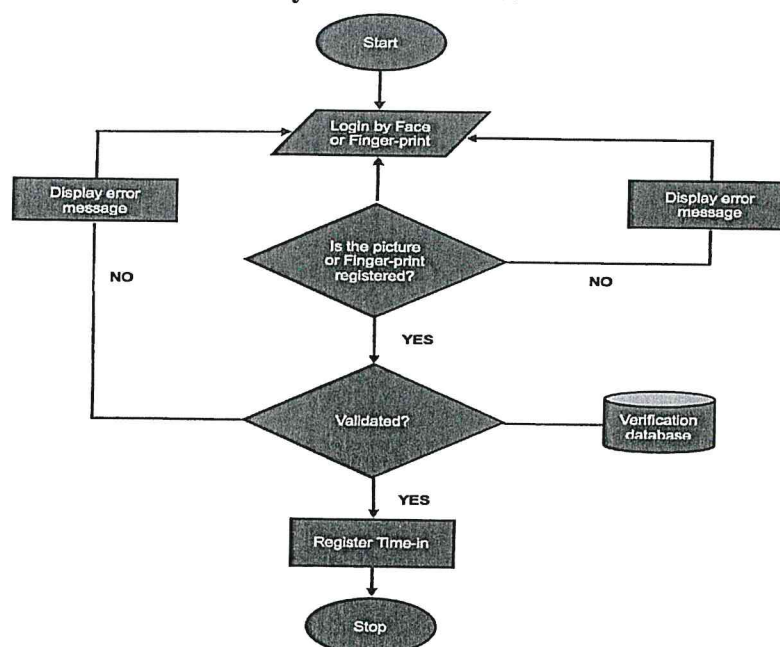
Mak-BAMS shall be implemented with centralized authentication through the electronic Human Resource Management System.

- i. The system shall enable staff to clock-in from any university installed attendance tracking biometric device on university premises to cater for the staff dynamic work schedules that go beyond their office stations to participate in university activities at different university premises

The Mak-BAMS allows employees clock-in offline and record attendance without an internet connection. After reconnecting to the internet, the system will synchronize the clock-in data with Mak-eHRMS.

- ii. The system enables staff to indicate periods of absence and the reason/ purpose (e.g., approved leave, official travel, fieldwork, research, community outreach, consulting work, scheduled/timetabled virtual teaching, etc) through the eHRMS.
- iii. The Mak-BAMS through the eHRMS, enables the newly established MakDATA integration platform to have automatic generation of reports and information on key functions including teaching and learning (e.g., student class attendance), onlinization, timetabling, and human resources, among others.

Fig. 1: Illustration of the Biometric System Flowchart



4. IMPLEMENTATION FRAMEWORK FOR THE MAK-BAMS

Mak-BAMS is implemented as a modular component of the Physical Identity and Access Management framework (Mak-PIAM), which incorporates Identity and Access Management to university premises and access to university information services and systems.

The system is implemented in synchronization with the Institutional Data Management Platform (Mak-DATA) for wholesome management of service performance and efficiency at Makerere University.

4.1. Biometric Devices

- i. The biometric devices support wireless connection, facial and fingerprint biometrics, and smart cards.
- ii. A biometric clock-in device is installed in the foyer/reception area of every single building at Makerere University's main and satellite campuses.

4.2. Biometric Data

- i. Biometric data is captured by the Directorate of ICT Support using a master clock-in biometric device. This device meets the global standards ISO 27001, ISO 38505, ISO 27701 and ISO 29151 ensuring secure storage and safe transmission of the captured data.
- ii. The captured data includes the staff biometric data and the attendance data. The biometric data is encrypted and includes the fingerprint, face and card. The Attendance data is processed and integrated with the existing university systems such as ACMIS, MUELE and e-HRMS among others.

5. USER GUIDELINES

This section provides a step-by-step guide to Makerere's biometric attendance management system, in response to what would constitute the frequently asked questions about the system.

5.1. Clocking-in

- i. The biometric clock-in devices will be configured to authenticate staff data against the e-HRMS database.
- ii. The biometric devices will be configured to permit clock-in at any time (beyond the regular 8am – 5pm) and any day of the week, thus accommodating the dynamic work schedules of staff at Makerere University.

- iii. The biometric clock-in devices are configured for online/offline mode, which enables staff to clock in even during periods of internet outage.
- iv. All university based biometric devices are synchronized which enables Staff to clock in from any approved device across the university premises at the Main or satellite campuses.

5.2. Staff off-campus on official duty.

- i. Staff intending to travel for official duties out of campus must seek and secure authorization as per the university's Human Resource Manual; and will thus, accordingly update the time away-for-official duties status through the eHRMS.
- ii. Authorization to travel must be done either through or updated in the e-HRMS of the university to enable synchronization with the Mak-BAMS.
- iii. The Mak-BAMS automatically records such staff as away-on-official duty from campus for the period indicated.

5.3. Online Teaching and Learning.

- i. The university's digital eLearning management platform (MUELE) is being integrated with the eHRMS and ACMIS systems, thus making it possible to extract attendance reports for online teaching.

5.4. Monitoring and Evaluation of Teaching activities.

- i. Mak-BAMS will be integrated with the ACMIS timetable system and SECAT to monitor and evaluate teaching activities.

5.5. Technical Support

- i. College IT personnel will be given preliminary training on basic first-line troubleshooting of the biometric devices to ably support staff at their units.
- ii. The Directorate of ICT Support will handle technical issues that the College IT personnel cannot troubleshoot.

6. DETAILED USER-GUIDELINES

The guidelines below have been developed to guide Makerere University staff on;

6.1. How to register for biometric attendance.

- i. The Records and Information Division of the Directorate of Human Resource will be responsible for capturing staff finger-print and facial-image biometric details.
- ii. In order to register for biometric attendance, a university staff will be required to present an appointment letter and a smart/RFID-enabled ID to a Records Officer at the Records and Information Division of the Directorate of Human Resource. These will be used to verify their identity before biometric data is captured.
- iii. New staff appointees will be required to register for biometric attendance within a period of two weeks after receiving their appointment letters from the Director Human Resource.
- iv. Before applying for a staff-ID, a new staff must create a university email address via <http://sso.mak.ac.ug/> . This email address will be embedded within the eHRMS system to enable initialization and access to electronic services including biometric access amongst others.
- v. To apply for a smart staff-ID, you must log into the eHRMS system <https://ehrms.mak.ac.ug/> and click-on “ID Requirements” under the menu on the left pen and read the guidelines for uploading both your ID picture and National ID.
- vi. Visit the Directorate of Human Resource for issuance of smart staff-ID after you have submitted requirements in (v) above.

6.2. How to revoke biometric attendance access privileges for staff.

The Directorate for ICT Support (DICTS) will revoke staff biometric access privileges upon receipt of a signed notice/letter from the Director Human Resource. This will be guided by the Amended (2009) Makerere University Human Resource Manual for cases where;

- i. Staff have died.
- ii. Staff have resigned or retired from university service.
- iii. Staff employment has been terminated.

6.3. **How to log tasks on the electronic Human Resource Management System (eHRMS)** <https://ehrms.mak.ac.ug/>

All university staff are required to log their weekly/monthly/quarterly or half year tasks on the eHRMS. This includes tasks such as workshops, conferences, seminars etc. scheduled to take place off-campus.

In order to log tasks on the eHRMS, you must log into the eHRMS system and;

- i. Click “My Tasks” under the menu on the left pen.
- ii. Click “New Task” to add a task and its duration.
- iii. You can add as many tasks as you wish.
- iv. Upon submission, your supervisor will be notified about your logged tasks via email and once approved, you will not be required to physically clock-in/out for tasks scheduled to take place off-campus.

6.4. **Best-user practices for the Biometric Attendance System.**

In order to ensure the longevity and proper functioning of biometric devices, users are required to adhere to the following best-practices.

- i. Staff are encouraged to minimize physical contact with biometric devices by clocking-in/out using facial recognition. This will increase the longevity of the devices and also reduce the likely transmission of disease-causing organisms arising from multiple fingers-prints touching the device.
- ii. Only clock-in/out using your finger-prints in instances where the biometric device is unable to recognize your facial-image. Ensure that your finger-print is not greasy, wet or dirty as this may affect the ability of the device recognizing your finger-prints. Also desist from pressing the device too hard as its sensors should be able to recognize you with only a light touch.
- iii. Log a ticket via <https://support.mak.ac.ug/> with subject “Biometric Failure” in instances where the biometric device is unable to recognize your finger-prints, facial-image or both and the Directorate for ICT Support (DICTS) will address your inquiry.
- iv. If you wear glasses with thick lenses, the facial recognition algorithm may struggle to accurately identify your face.
- v. The installed biometric devices are IP65 compliant (resistant to water and liquid splashes). Nonetheless, staff are advised to desist from wetting the devices especially with alcohol-based sanitizers which may affect the devices thumb-

- print scanner platen (polycarbonate glass-plate where you place your finger-print).
- vi. DICTS will periodically assess the security and functionality of all biometric devices in the university to ensure optimal performance.

6.5. **Biometric Data Security and Privacy**

The Makerere University Data Security policy broadly classifies university data into Public, Private and Restricted Data. Staff and Student biometric data (finger-prints, facial-images, iris, voice-waves etc.) is classified as restricted-data whose collection, storage, usage and disposal must adhere to a strict set of procedures in order to ensure its Privacy, Integrity and Security.

The security guidelines below, relate to the collection, usage, storage and disposal of staff biometric data. The guidelines are aligned to the Makerere University Data Security Policy, the Uganda Data Protection and Privacy Act, 2019, the Data Protection and Privacy Regulations, 2021 and the guidelines of the National Data Protection Office

i. **Collection of staff biometric-data.**

- Staff biometric data shall be collected by the Records & Information division of the Directorate of Human Resource (DHR).
- The Records & division shall only capture staff facial-image and finger-prints biometrics.
- Staff shall sign a “biometric data capture form” consenting to the capture of their personal data. The form can be downloaded from the Human Resource Management System <https://ehrms.mak.ac.ug/> , filled & signed-off by hand, scanned and uploaded to the Human Resource Management System.

ii. **Storage of staff biometric-data**

- Staff biometric and attendance data shall be stored centrally in a secure & encrypted password protected database in the attendance management system. Encrypted data cannot be used to recreate the complete original image.

User-Access is only available to the attendance data. In order to ensure maximum security, finger-print data cannot be accessed by any user and is therefore encrypted. This kind of data can only be changed by taking the staff biometrics afresh.

- All staff attendance and biometric data is securely backed-up on a daily basis following an automated back-up procedure.

- Entry to data-centers that host Makerere University Information Systems is restricted through physical security and biometric access. Only authorized IT personnel can access these facilities.
- iii. **Usage of staff biometric data**
 - Staff biometric data will strictly be used for attendance management (clocking-in/out).
- iv. **Ethical compliance**
 - Makerere University shall not sell, lease, trade, or otherwise profit from staff biometric identifier or information.
 - Makerere University shall not disclose, re-disclose, or otherwise disseminate staff biometric identifier or information unless required by law.
 - Makerere University shall not share personal information; including, but not limited to fingerprint information, with any non-vendor third party organization.

6.6. **Staff biometric data disposal**

- The Directorate for ICT Support (DICTS) shall disable staff biometric access upon staff death, resignation, retirement or termination.
- Biometric data for staff who have died, resigned, retired or been terminated will be securely stored and only retrieved to aid investigations.
- Biometric data for deceased staff shall be destroyed in accordance with the Laws governing data retention prevailing at the time.

6.7. **Locations and physical security of biometric devices**

- i. The biometric devices are installed in entrance-ways/foyer or reception areas of buildings which makes them accessible and visible to all.
- ii. Custodians and security personnel of university premises as well as IT personnel at user-units (colleges), are required to ensure the physical security of biometric devices at their units.
- iii. The Directorate for ICT Support (DICTS) will continuously monitor the up-time/availability of all biometric devices via an electronic console/system and will notify college IT personnel should any biometric device be unavailable.

- iv. The biometric devices are tamper-proofed with an alarm system which will be triggered in the event that the device is forcefully tampered with.
- v. The biometric devices are equipped with a 180-degree view camera that will automatically capture photos of anyone tampering with the device.
- vi. The security office of the university will ensure the security of all biometric devices installed at the main-gate, western-gate and eastern-gates of the university.

6.8. Review and update of the biometric attendance guidelines

- i. The Directorate of Quality Assurance will analyze staff biometric attendance data on a quarterly basis for purposes of identifying areas of improvement as well as ensuring compliance with university policies of teaching, learning and research amongst others.
- ii. The Biometric Attendance Guidelines will be reviewed twice every year by a committee comprising of membership from the Directorate of Human Resource, Quality Assurance, the Academic and Administrative staff associations as well as DICTS. The Biometrics Review Committee will be chaired by the Deputy Vice Chancellor in charge of Academic Affairs.
- iii. The Biometrics Review Committee will submit their recommendations to the Legal, Rules, Privileges and Estates Committee for review and approval.

6.9. End-user support guidelines

i. 1st Line Technical Support

If the biometric device at your user-unit is not powered, report the incident to your college IT personnel. This may be a local power-related issue that the user-unit IT personnel can resolve. If all fails, they will log a ticket via <https://support.mak.ac.ug/> and escalate the inquiry to DICTS.

ii. 2nd Line Technical Support

If the biometric device is unable to recognize your facial-image, finger-prints or both, first refer to the best-user best practices for the biometric attendance system for guidance. If all fails log a ticket via <https://support.mak.ac.ug/> for DICTS to address your inquiry.

iii. **3rd Line Technical Support**

In the event that the biometric hardware or software fails, DICTS will contact the manufacturer for support. In such circumstances, staff will be advised to clock-in/out from any device they can access including those at university gates as they await repair of device(s) for their premises.

Approved by the University Council this ^{25th} day of July...2024



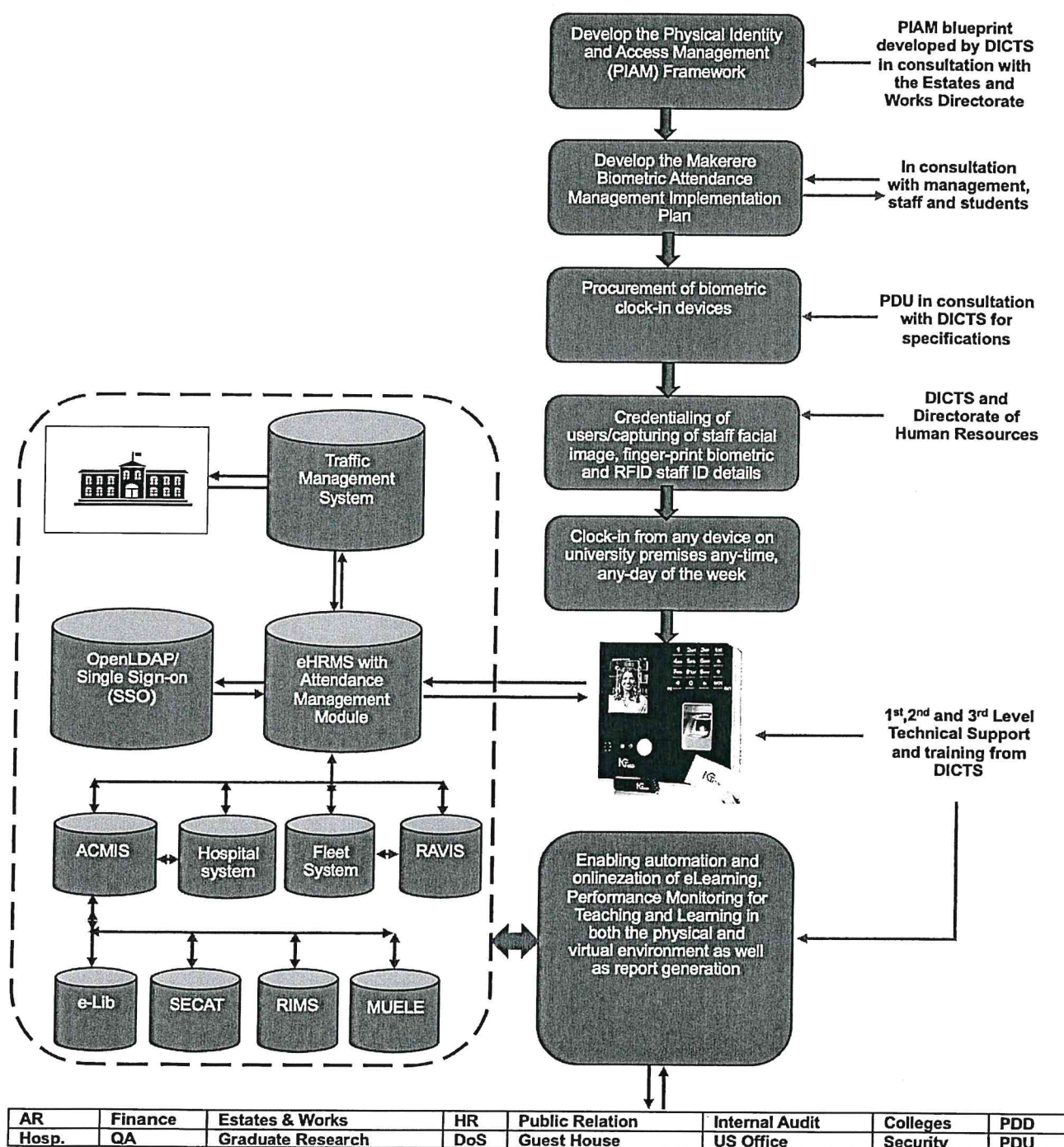
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Lorna Magara (Mrs.)
CHAIRPERSON COUNCIL



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Yusuf Kiranda
UNIVERSITY SECRETARY

APPENDICES

A: Detailed Architectural Description of the Mak-BAMS



Proposed Breakdown of Academic Staff Workload

Workloads Description & Distribution	Hours Per Week (Min/Max)	Expected Work Activities and Output/ Indicators
1. Teaching Contact Hours	10 -12	• Contact hours with students in conducting Lectures, Tutorials, Clinical work, Laboratory practical work and Fieldwork. • Administering Coursework and Progressive Assessments: Tests, Quizzes, Presentations, Take-home assignments and assessment. • Final examinations setting, moderation, invigilation, marking and grading.
2. Teaching Preparations	4 – 6	• Course contents design, Curriculum review, Lecture planning, Library hours, Teaching resources and material preparations. • Updating course contents and materials on learning management system (MUELE).
3. Students Mentorship and Projects Supervision	6 – 8	• Providing students consultation hours and career mentorship. • Supervising students research and academic projects. • Monitoring and supervision of students' Internships, Field-attachment, Industrial training. • Marking dissertations and theses. • Attending graduate viva-voce and defense.
4. Research and Publication	10 -12	• Staff research proposals and Grants writing, • Conducting research fieldwork and desk-review. • Production of research publications (books, monographs, articles), • Conducting Internal research seminars and lecture series.
5. Outreach and Community Engagement	4 – 6	• Community training and capacity building, • External dissemination workshops and seminars, • Executing strategic partnership and MoUs with governmental and non-governmental agencies, • Participation in community welfare and livelihood development projects, • Participation in external academic and professional vetting, reviews and examinations, • Membership on Boards of professional bodies, Governance Boards and Councils.
6. University Administrative Work	4	Department/College meetings, Administration, Administrative Assignments, Vivas

NOTES:

- i. All Staff will be required to submit to their respective heads and supervisors, their individual workload plans and activity distribution, following the workers hours as per the policy.